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CLIMATE SOLUTIONS

Workers keep dying from heat. Data from inside their bodies shows why.



PHILADELPHIA, PA - AUGUST 4: Construction worker pours water on his face to cool off as he digs a sanitation pipe ditch during a heatwave on August 4, 2022 in Philadelphia, Pennsylvania. (Photo by Mark Makela/Getty Images) (Mark Makela/Getty Images)

By [Nicolás Rivero](#) and [Niko Kommenda](#)

Today at 6:00 a.m. EDT

Six hours before she collapsed from the heat in the South Florida tropical plant nursery where she works, Irma wasn't feeling well.

By 10 a.m., two hours into her shift weeding long rows of plants on a hot August day, her head hurt and she felt nauseous. By 2 p.m. she had full body cramps and chills. At her boss's urging, she pushed through the pain without a break until her body gave out at 4 p.m. Her vision went black and she fell to the ground.

Irma asked to be identified only by her first name for fear of retaliation from her employer. Her boss still won't let her or her co-workers pause to rest on hot workdays, she said, aside from a half-hour break for lunch at noon.

"I'm bent over all day working," she said. "If you stand up, the bosses ask, 'Are you feeling well?' But if you say, 'No, I don't feel well,' they'll send you home and you'll lose a day of work."

Incidents such as Irma's are a life-threatening symptom of dangerous working conditions in a country that has no national heat protections for workers. Data collected by a team of researchers led by Roxana Chicas, an assistant professor at Emory University's Nell Hodgson Woodruff School of Nursing shows how some laborers face danger on a frequent basis — and how periodic cooling breaks can prevent their bodies from getting dangerously hot.

Heat kills dozens of U.S. workers and injures thousands more each year according to Labor Department figures, although a 2024 report acknowledged official data "vastly undercount cases of heat-related injuries and illnesses among workers."

There is no nationwide rule requiring that employers provide paid breaks on hot days. Seven states have passed such laws, but a federal rule begun under the Biden administration now faces an uncertain future under President Donald Trump.

The World Health Organization and the U.S. government say people should avoid pushing their core body temperature beyond a feverish 100.4 degrees for extended periods. But data Chicas and her colleagues collected from Florida agricultural workers show they routinely pass that threshold on the job.

The scientists asked workers to swallow a pill-sized thermometer and to use an activity monitor.

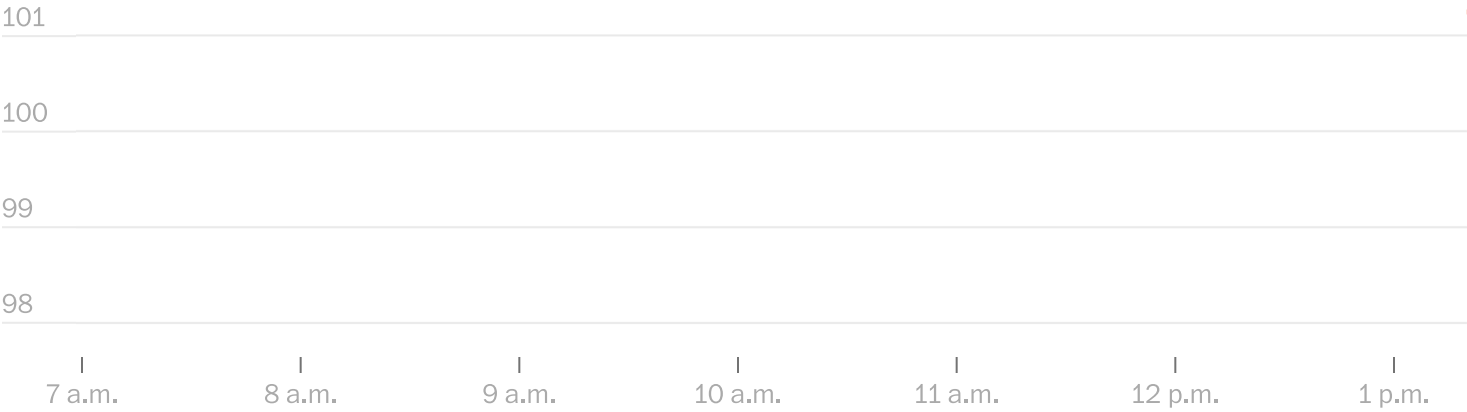
These sensors measured the core body temperature and the activity level of 221 workers every 30 to 60 seconds.

Over the course of an average work day, **49%** reached a core body temperature of at least 100.4F — similar to a fever from an infection.

Core body temperature

103F
102

Here is a closer look at the readings from one worker, a 39-year-old woman picking crops in Immokalee, Florida.



Her **body temperature** is around 98F at the start of her shift but soon starts rising.

The **activity monitor** strapped to her waist shows constant movement, reflecting the fast-paced field work.

The only time the worker's body temperature drops is when she takes a **half-hour break** around noon. Nevertheless, by the time she clocks out at 2:30 p.m., her temperature has reached 102.7F.

Everybody reacts to heat differently, and not every worker will see such a large drop in their body temperature when they take a break, according to Lee Newman, a physician and professor at the Colorado School of Public Health. He also stressed that 100.4 degrees isn't a magic switch that takes a worker from zero risk to full-on heatstroke. But, generally, as workers' temperatures rise, bad things start happening inside their bodies.

Overheated workers sweat and may get dehydrated. Their hearts work harder to pump more blood into their straining muscles and to their skin to cool off, while shifting blood flow away from organs such as the kidneys, stomach and intestines. Workers can become lightheaded, limiting their ability to think clearly. Inflammation can leave their tissues more susceptible to damage. Overworked and oxygen-starved cells may begin to break down and die.

Most workers won't get so sick they collapse or have to go to the hospital. But bit by bit, the strain of heat and dehydration chips away at their kidneys and other organs, leading to long-term health problems.

"Every day they're getting a little hit to their kidneys, and over time they run the risk of chronic kidney failure," said Newman.



Construction workers on a project in Clarksburg, Maryland, do roof work on a 90-degree day. (Michael S. Williamson/The Washington Post)

Taking a break, even for 10 or 15 minutes, disrupts that process. When you work hard, your own body becomes a major source of heat. Resting prevents your body from heating itself up more, and it gives you time to actively cool yourself down by getting into shade, standing in front of a fan or putting a cold, wet towel on your body.

“It also gives people a good opportunity to rehydrate, because sometimes it’s hard to drink a decent amount of water while you’re doing your work,” said Douglas Casa, head of the University of Connecticut’s Korey Stringer Institute, which publishes research on heat illness among workers and athletes.

Regular breaks are especially important for laborers because they typically spend more time, day after day, exerting themselves in the heat than athletes, gardeners and those trying to exercise, said Zachary Schlader, an associate professor of kinesiology at the Indiana University School of Public Health-

Bloomington. Workers may also be older and have more health complications, which raises the risk something will go seriously wrong.

“It’s the one-in-10,000 person at work who has the bad day,” said Casa. “That bad day might happen because they had a gastrointestinal illness the last couple of days and they were dehydrated, they had nausea, they had diarrhea. Or they’re just coming off the flu, or they’re coming back from four weeks off because they had broken their ankle, or they were rotated from North Dakota down to New Mexico and it’s a new stress.”

A proposed federal rule would require workers to take 15-minute breaks every two hours when the heat index rises above 90 degrees. That’s “better than not having any plan,” Schlader said, but the length and frequency of breaks workers need to avoid health problems varies a lot depending on the work, the weather, how acclimated workers are to the heat and the gear they’re wearing.

For lighter work in moderate heat, a few minutes every few hours might be plenty. For workers such as Polymice, a South Florida nursery worker who sprays pesticides over acres of plants while wearing a respirator mask and a head-to-toe protective plastic suit, breaks should be much longer.

“When it gets hot, you just start to sweat, sweat, sweat. I sweat so much that it runs down into my gloves and starts to fill them up,” said Polymice, who also asked to be identified by his first name to avoid trouble at work. “You take the suit off, you rest for a bit, and then five minutes later you put the suit back on and you’re right back in the same heat.”

It’s also important that workers get paid during their breaks — or, at least, that they can keep working after a break to make a full day’s wages. Otherwise, they’ll be discouraged from resting.

On the day Irma had a heatstroke last August, when she passed out 30 minutes before the end of her shift, her boss sent her to an air-conditioned office to cool

off and drink water until quitting time. But when her paycheck came at the end of the week, Irma realized the boss had docked her half an hour's pay.

In the absence of federal rules, Polymice and Irma are both organizing their fellow nursery workers to push their bosses for better working conditions, including breaks on hot days, as part of a campaign called Planting Justice. Organizers say they hope to pressure tropical plant nurseries in South Florida to change their ways by encouraging American houseplant customers to boycott companies that mistreat workers — a strategy that Florida tomato workers used decades earlier to create some of the strongest heat protections in the country.

“We want shade, water, rest and medical protections,” Irma said. “We’re fighting for the rights we deserve.”

The silhouettes of people shown in the interactive graphic are from ProPublica’s WeePeople typeface.

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